

CHEN XU

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COMPETITIONS

International:

- **ICPC** (International Collegiate Programming Contest) Regional Contest: **Gold Medal**, 2025
- **ICPC** Asia East Continent Final: **Silver Medal**, 2024
- **Hackathon** Shipyard2026 Winner
- **Codeforces International Master**: [Codeforces Profile](#)

National:

- **USACO** Certified: **Platinum Division**, 2025

TECHNICAL EXPERIENCE

Computer Vision Engineer | Golden Stone Technology | May - June 2025

- Developed facial recognition and liveness detection system by training machine learning models to identify phone screen reflections for distinguishing real humans from photos/videos in industrial safety applications

Computer Vision Engineer | Golden Stone Technology | June - September 2025

- Developed coal rock detection and separation algorithm. Uses self-trained Machine Learning models and OpenCV. Overall accuracy of 97%.

Software Engineering Intern | Sohu | July 2025

- Designed an algorithm to auto-categorize legacy videos (5,000+ files), improving retrieval efficiency by 7%

PROJECTS

BookaCook - iOS Application | 2026 | 100+ Downloads | Hackathon Winner (\$20,000 Prize)

- Built and shipped a full-stack iOS recipe platform backed by SQLite; developed Python web crawlers to scrape, deduplicate, and index 1,200+ cookbooks and 1M+ recipes; integrated ISBN barcode scanning to instantly catalog physical cookbooks into the user's library
- Engineered multi-model AI pipeline: TikTok API ingestion → Groq Whisper speech-to-text → GPT-4o vision for ingredient recognition → Groq Llama 3 8B Instant for structured recipe generation; combined with hybrid collaborative-filtering recommendation system for personalized discovery

RightToRecord - iOS Application | March - April 2025 | 80+ GitHub Stars. [GitHub Link](#)

- Developed civic technology app using Swift, Python, and AWS for secure cloud deployment
- Implemented real-time video upload system to protect recordings from device confiscation
- Reached 80+ active users; received testimonials confirming successful evidence preservation

Facial Recognition & Liveness Detection | 2025 | 70+ GitHub Stars. [GitHub Link](#)

- Trained machine learning models to detect phone screen reflections on faces, enabling real-time distinction between live persons and photo/video spoofing attempts
- Open-sourced complete system with comprehensive documentation, demonstrating practical anti-spoofing techniques in facial recognition applications

Large Language Model (LLM) from Scratch | 2024

- Built custom language model architecture from the ground up to understand transformer mechanics and computational constraints
- Explored hardware-software optimization challenges in deep learning and investigated resource-efficient training methods

TECHNICAL SKILLS

C++, Python, Swift, OpenCV, ML, Computer Vision, Deep Learning, AWS, Git

EDUCATION

Computer Science Class 2030. New York University

Languages: English & Mandarin